## FRONT COMBINATION LIGHT

id091800010200

### Purpose

### Bulb type

- The front combination light groups and houses together the parts related to the front exterior lights such as the headlight, parking light, front turn light, headlight leveling actuator (with headlight manual leveling system), wiring harnesses, and connectors.

### LED type

- The front combination light groups and houses together the parts related to the front exterior lights such as the headlight, parking light, front turn light, headlight leveling actuator, wiring harnesses, and connectors.
- Energy consumption reduction and maintenance-free design been achieved with the adoption of LEDs on the headlights LO and HI.
- Visibility of road signs positioned higher up has been improved by blurring the cut lines of the headlights between light and dark to the extent possible.
- Visibility of pedestrians and roadside zones has been improved by expanding the light distribution of the headlights.

### Function

- Each light turns on or flashes when the light switch, turn switch, or hazard warning switch is operated.

### Construction

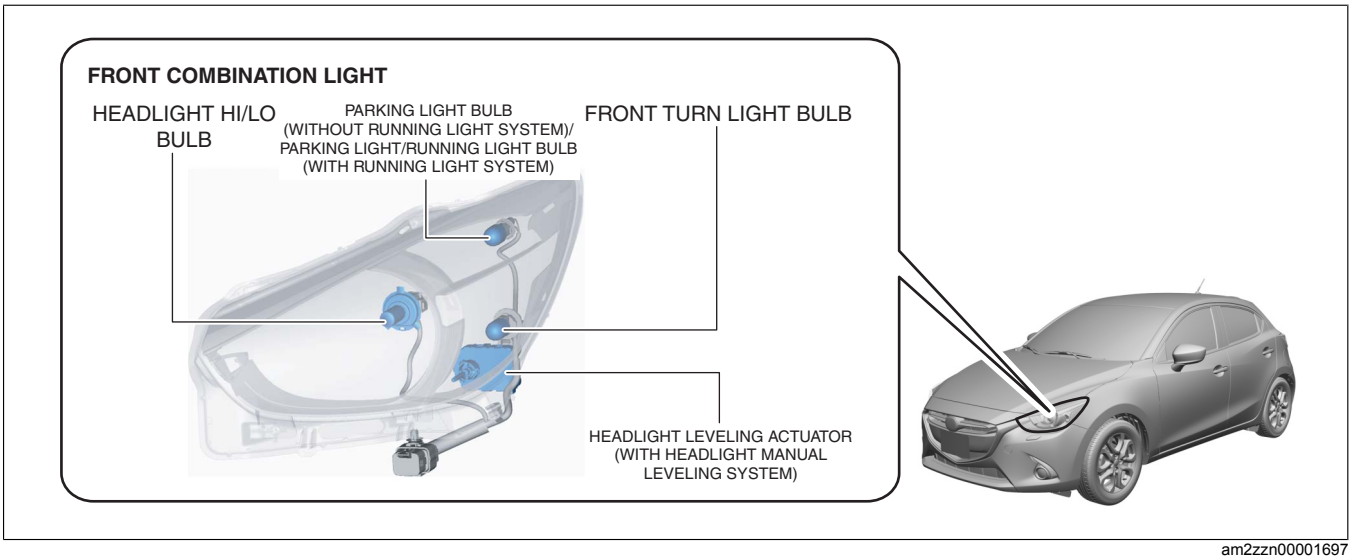
- A bulb type or LED type is available for the front combination light.

### Bulb type

- The front combination light groups the following parts:
  - — Headlight HI/LO bulb
  - — Parking light bulb (without running light system)
  - — Parking light/running light bulb (with running light system)
  - — Front turn light bulb
- The front combination light is integrated with the headlight leveling actuator. (with headlight manual leveling system)

**Note**

- Fogging or condensation may occur inside the front combination lights, however, it is a natural phenomenon occurring as a result of a temperature difference between the interior and exterior of the front combination lights and it does not affect the light performance.
- Fogging or condensation occurring as a natural phenomenon will dissipate when the temperature inside the front combination lights rises after the headlights are turned on and a period of time has elapsed.




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### LED type

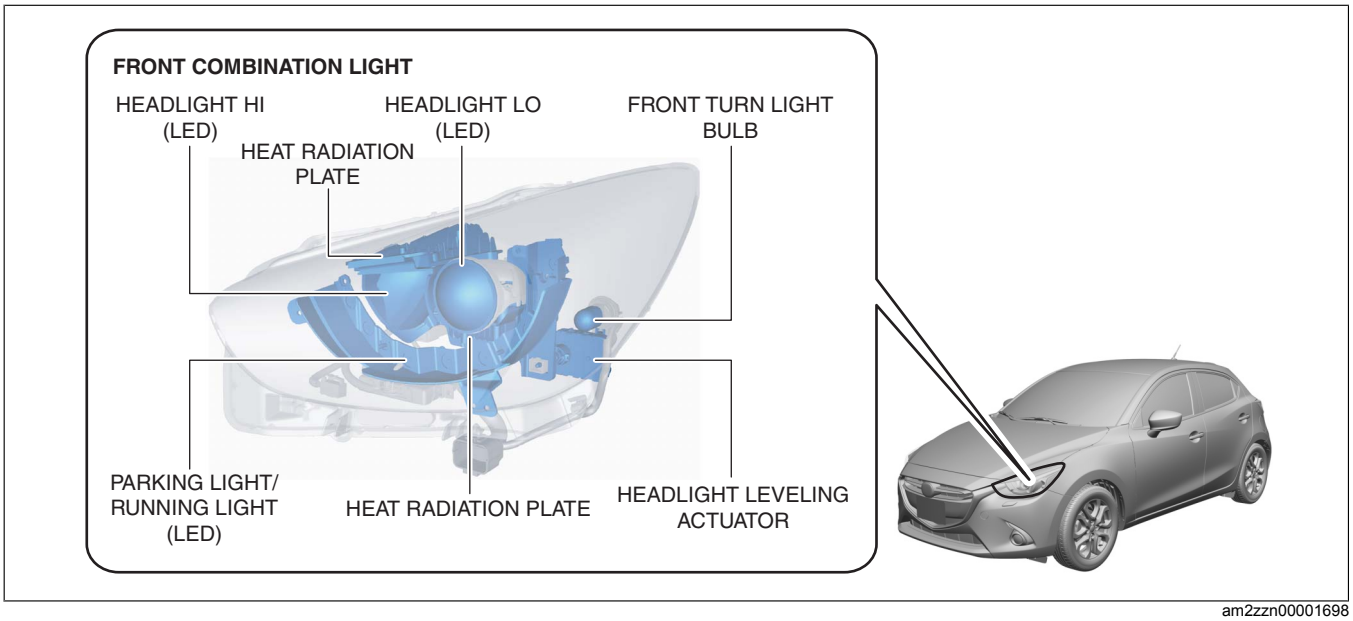
- The front combination light groups the front turn light bulb.
- The front combination light is integrated with the following parts:
  - — Headlight LO (LED)
  - — Headlight HI (LED)

— Parking light/running light (LED)
— Headlight leveling actuator

- A projector type headlight has been adopted to the headlight LO.
- Heat generated by the illumination of the LEDs is absorbed and dispersed by the heat radiation plates installed to the back of the headlight LO and HI reflectors.

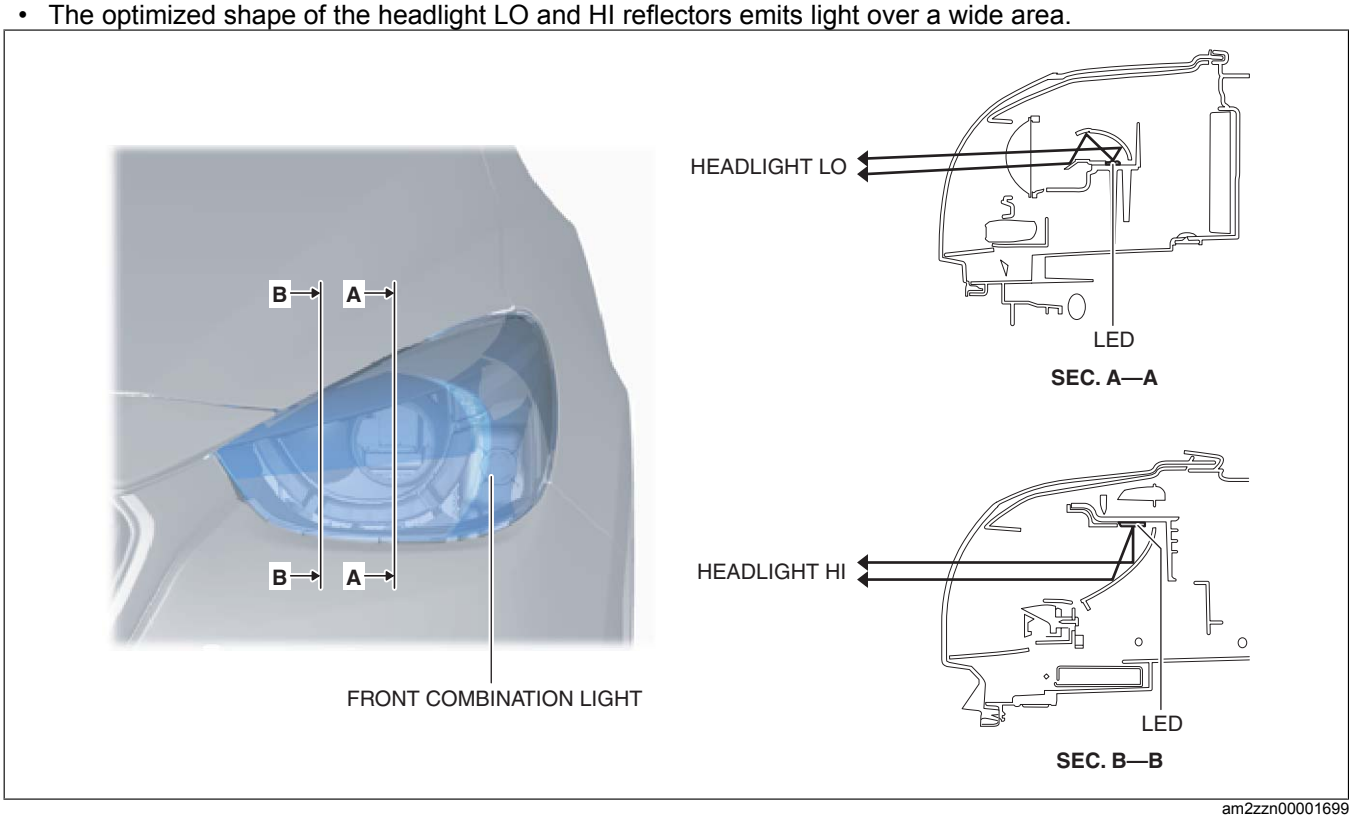
**Note**

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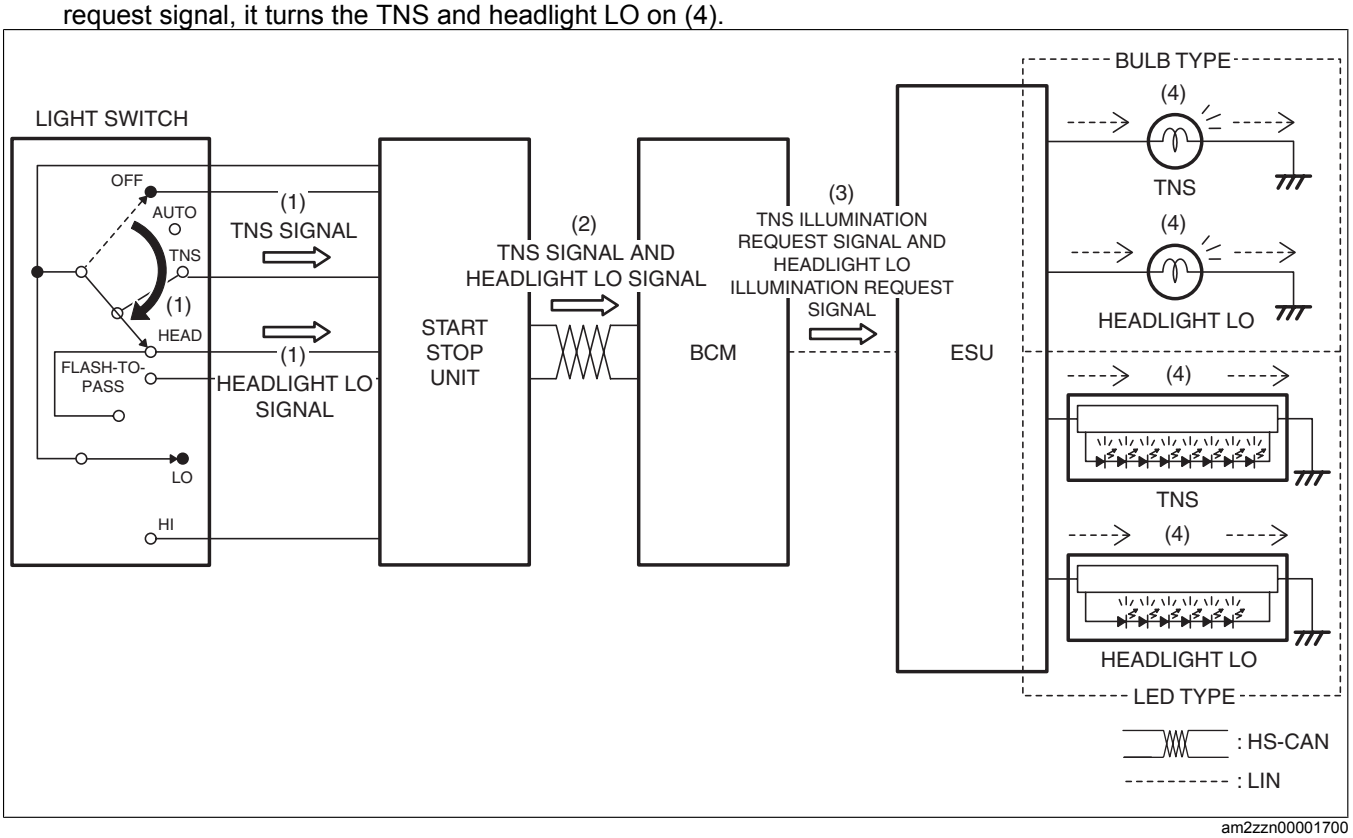
- The optimized shape of the headlight LO and HI reflectors emits light over a wide area.




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**Operation**
**Headlight LO**

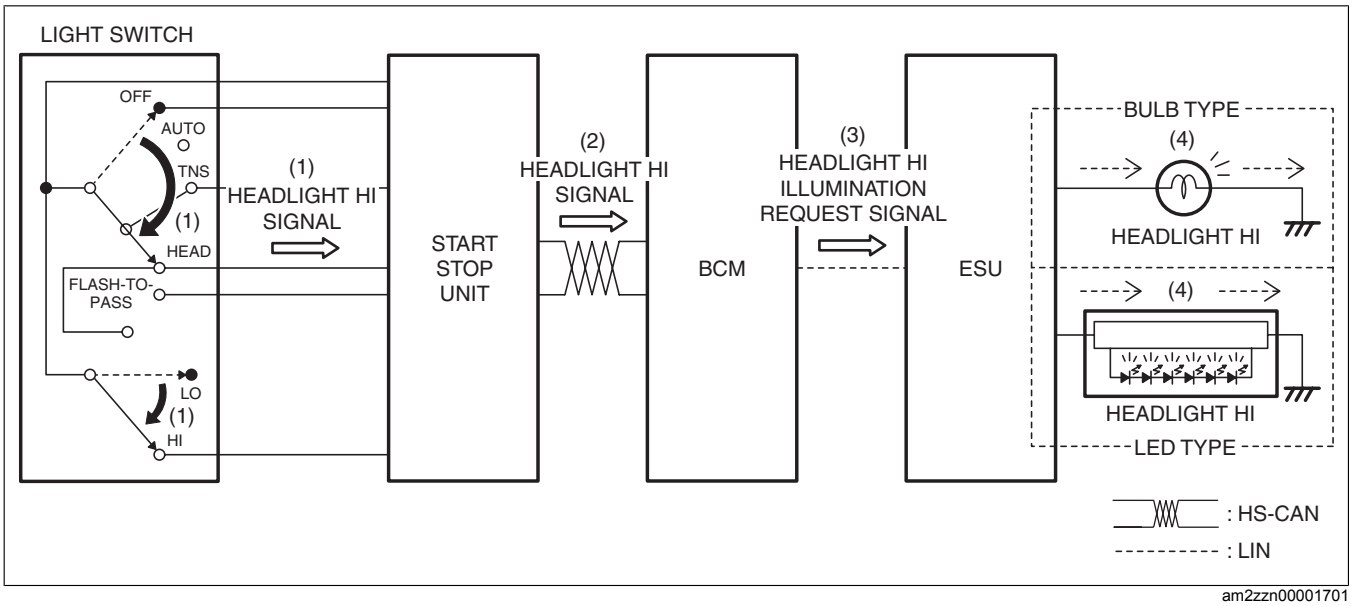
1. When the light switch is moved to the HEAD position and the dimmer switch is moved to the LO position, the start stop unit detects (1) a TNS signal and headlight LO signal.
2. The start stop unit sends (2) the TNS signal and headlight LO signal to the BCM via CAN communication.
3. When the BCM receives the TNS signal and headlight LO signal with the ignition switched ON (engine off or on), it sends (3) a TNS illumination request signal and headlight LO illumination request signal to the electrical supply unit (ESU) via LIN communication.
4. When the electrical supply unit (ESU) receives the TNS illumination request signal and headlight LO illumination request signal, it turns the TNS and headlight LO on (4).




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**Headlight HI**

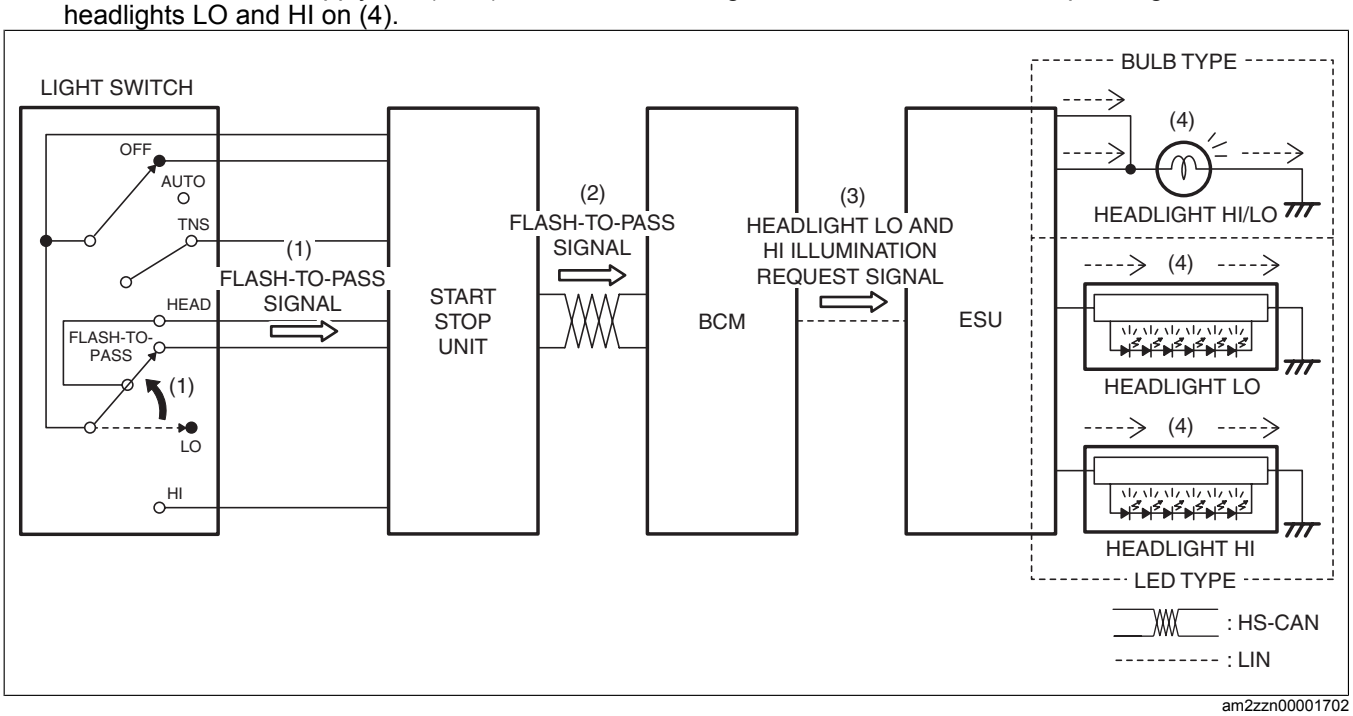
1. When the light switch is moved to the HEAD position and the dimmer switch is moved to the HI position, the start stop unit detects (1) a headlight HI signal.
2. The start stop unit sends (2) the headlight HI signal to the BCM via CAN communication.
3. When the BCM receives the headlight HI signal with the ignition switched ON (engine off or on), it sends (3) a headlight HI illumination request signal to the electrical supply unit (ESU) via LIN communication.
4. When the electrical supply unit (ESU) receives the headlight HI illumination request signal, it turns the headlights HI on (4).




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**Flash-to-pass**

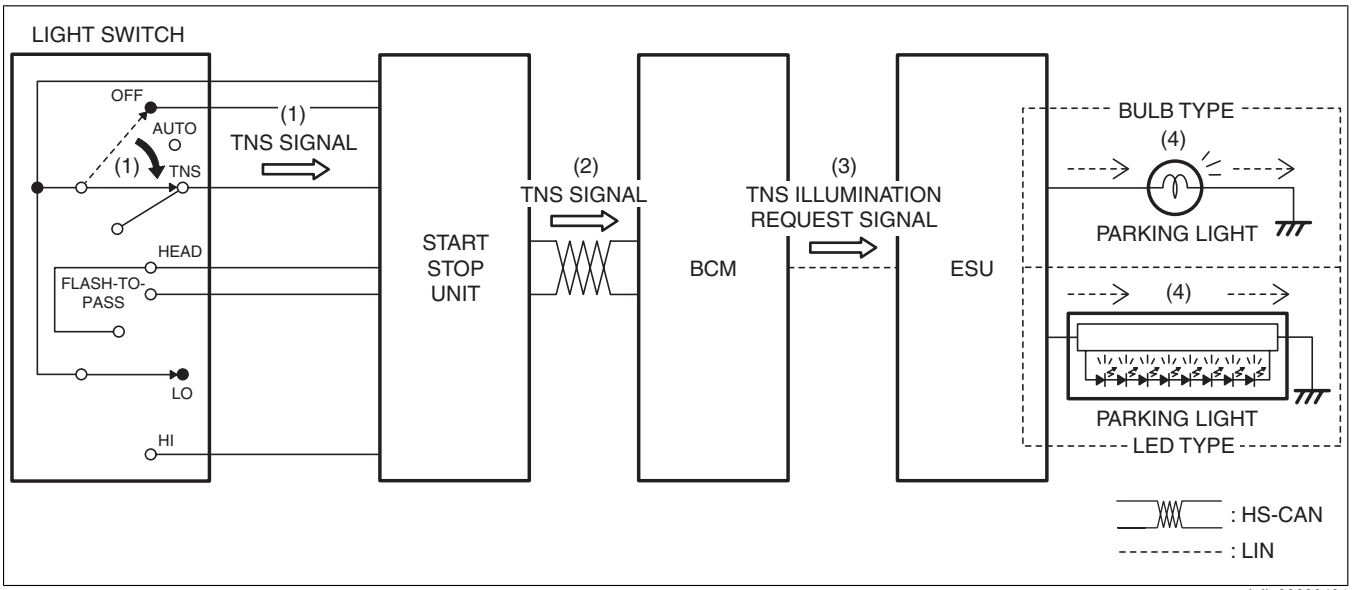
1. When the dimmer switch is moved to the FLASH-TO-PASS position, the start stop unit detects (1) a FLASH-TO-PASS signal.
2. The start stop unit sends (2) the FLASH-TO-PASS signal to the BCM via CAN communication.
3. When the BCM receives the FLASH-TO-PASS signal with the ignition switched ON (engine off or on), it sends (3) headlight LO and HI illumination request signals to the electrical supply unit (ESU) via LIN communication.
4. When the electrical supply unit (ESU) receives the headlight LO and HI illumination request signals, it turns the headlights LO and HI on (4).




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**Parking light**

1. When the light switch is moved to the TNS position, the start stop unit detects (1) a TNS signal.
2. The start stop unit sends (2) the TNS signal to the BCM via CAN communication.
3. When the BCM receives the TNS signal, it sends (3) a TNS illumination request signal to the electrical supply unit (ESU) via LIN communication.
4. When the electrical supply unit (ESU) receives the TNS illumination request signal, it turns the parking lights on (4).

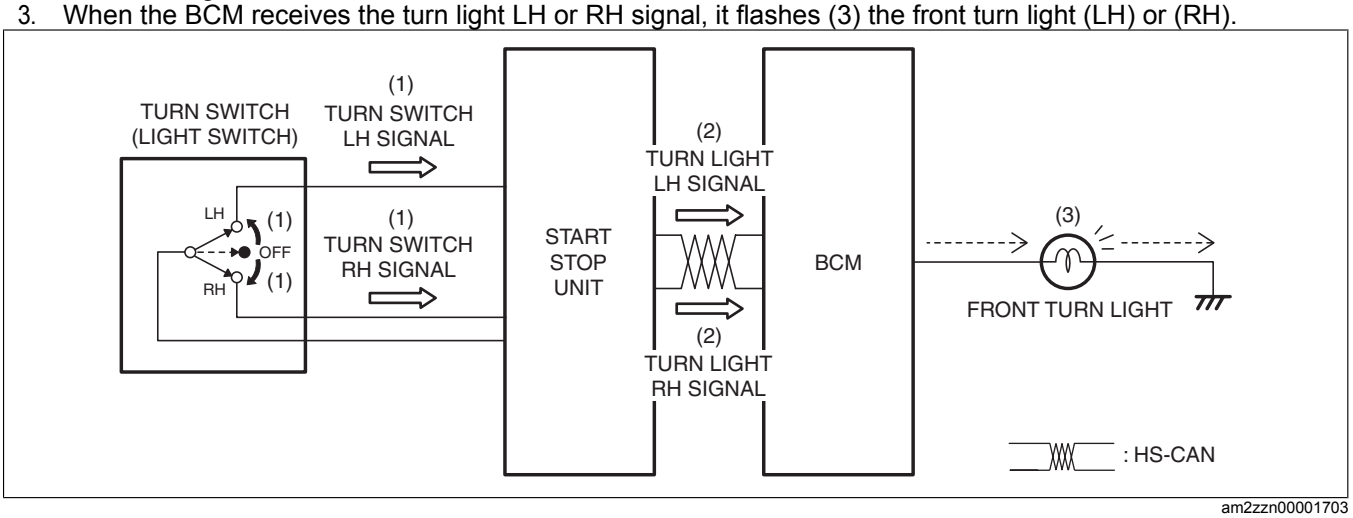



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**Front turn light**

**Turn light system**

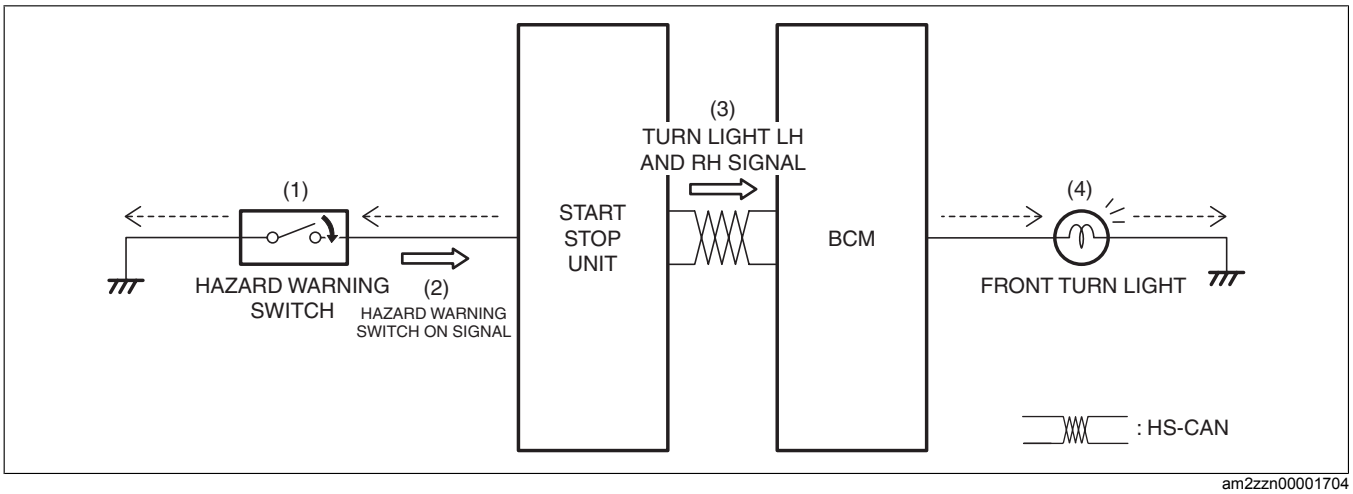
1. When the turn switch is moved to the LH or RH position with the ignition switched ON (engine off or on), the start stop unit detects (1) a turn switch LH or RH signal.
2. The start stop unit sends (2) the turn switch LH or RH signal to the BCM via CAN communication as a turn light LH or RH signal.
3. When the BCM receives the turn light LH or RH signal, it flashes (3) the front turn light (LH) or (RH).




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**Hazard warning system**

1. When the hazard warning switch is turned on (1), the start stop unit detects (2) a hazard warning switch on signal.
2. The start stop unit sends (3) the hazard warning switch on signal to the BCM via CAN communication as a turn light LH and RH signal.
3. When the BCM receives the turn light LH and RH signal, it flashes (4) the front turn lights (LH) and (RH).



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**Headlight leveling actuator**

- Refer to the [HEADLIGHT LEVELING ACTUATOR]. (See HEADLIGHT LEVELING ACTUATOR.)

**Fail-safe**

- Not applicable